

# NBA Draft Prospect Prediction - ML Project Overview

## Project Overview

The goal of this machine learning project is to determine whether or not a player from the global basketball players' statistics dataset is an NBA draft prospect.

The dataset contains detailed player statistics from leagues all over the world, including data on NBA players and those who successfully transitioned from international leagues to the NBA.

Success is defined as correctly identifying players who can make the transition to the NBA. The objective is to use player statistics to build a model that can predict-with at least 95% confidence-whether a player has NBA potential.

This project follows a structured pipeline with clearly defined milestones that must be completed to achieve a successful model.

## Milestones

1. Data Preparation
2. Exploratory Data Analysis (EDA)
3. Data Cleaning
4. Feature Engineering
5. Feature Selection
6. Model Selection & Handling Imbalanced Data
7. Model Fine-Tuning

## Data Preparation

- Text Cleaning: `clean_text(text)`
- Data Transformation: Extracted `birth_year` and `birth_month` from `birth_date`, then dropped `birth_date`
- Data Type Conversion: Converted columns to string types as needed
- Column Removal: Dropped `height`, `weight`, `REB`, `PTS`, `draft_pick`, `draft_round`
- Feature Creation: Created `draft_position` and `NBA` columns
- Category Reduction: Reduced cardinality in `League` and `nationality`

## Exploratory Data Analysis (EDA)

- Followed EDA protocol
- Generated descriptive statistics using `AutoViz`
- Analyzed the `NBA` target column
- Explored categorical data
- Plotted histograms for continuous variables
- Used skewness and heatmaps for distribution and correlation
- Created T-statistic visualizations

## Data Cleansing

- Outlier detection and visualization using the IQR method
- Missing value imputation using `KNNImputer`

## Feature Engineering

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- Created BMI column
- Combined regular season and playoff stats per NBA player
- Visualized NBA draft sources and player positions

### **Feature Selection**

- Used Lasso, Ridge, LinearSVC, GradientBoostingClassifier, and RandomForestClassifier
- Dropped League and Stage due to perfect correlation with NBA

### **One-Hot Encoding**

- Completed during earlier stages of data preparation

### **Model Selection & Fine-Tuning**

- Evaluated models using Accuracy, Precision, Recall, and F1 Score
- Classifiers: LogisticRegression, DecisionTreeClassifier, RandomForestClassifier, GradientBoostingClassifier, AdaBoostClassifier, SVC, KNeighborsClassifier, GaussianNB, XGBClassifier
- Addressed class imbalance using SMOTE
- Fine-tuned top models: XGBClassifier and GradientBoostingClassifier using GridSearchCV